

Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 6 – Rotation and ω

Where the SHM equations come from

IB Physics Guide • §C.1 Simple harmonic motion • Theme C, Wave Behaviour

This worksheet will cover the following topics:

C1-2 the defining equation $a = -\omega^2 x$ **C1-3** describing an oscillator by period, frequency, angular frequency, amplitude, equilibrium position and displacement

C1-4 $T = \frac{1}{f} = \frac{2\pi}{\omega}$

C1-9 $x = x_0 \sin(\omega t + \phi)$, $v = \omega x_0 \cos(\omega t + \phi)$, $v = \pm \omega \sqrt{x_0^2 - x^2}$

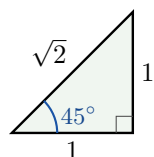
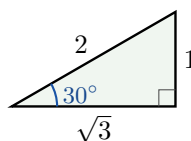
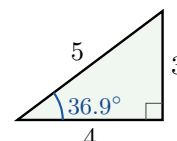
Linking questionHow can circular motion be used to visualize SHM? (*links to A.2*)

Background Knowledge

Read this as you come in. Highlight the key words, and use the space to add your own notes.

A. Three triangles worth knowing by heart

Three right-angled triangles turn up so often that their sides are worth memorising. Each one gives you exact values of sin and cos with no calculator.

**Half a square.** Two equal sides, so the angles are 45° and 45° . $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.71$ **Half an equilateral triangle.** Sides 1, $\sqrt{3}$, 2, angles 30° and 60° . $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2} \approx 0.87$ **The 3–4–5 triangle.** $3^2 + 4^2 = 5^2$, so the sides are whole numbers. $\sin 36.9^\circ = \frac{3}{5}$, $\cos 36.9^\circ = \frac{4}{5}$

In every right-angled triangle:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

B. Cos is the x, sin is the y

Put a right-angled triangle inside a circle of radius 1, with one corner at the centre. The hypotenuse is the radius, so it is 1, and the two other sides are simply

$$x = \cos \theta \quad y = \sin \theta$$

cos measures how far across, **sin** how far up.

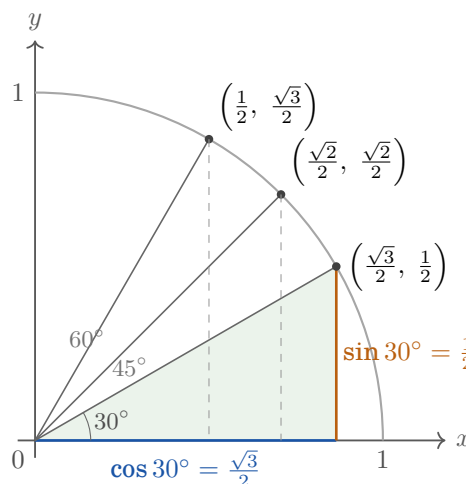
For a circle of radius r , every length scales by r :

$$x = r \cos \theta \quad y = r \sin \theta$$

Example. $r = 5$ at 36.9° gives $x = 5 \times \frac{4}{5} = 4$ and $y = 5 \times \frac{3}{5} = 3$: the 3-4-5 triangle again.

angle θ	0°	30°	45°	60°	90°
sin θ (the y)	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos θ (the x)	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

Read along the rows: sin climbs from 0 to 1 as cos falls from 1 to 0. They are the same numbers in the opposite order, because $\sin \theta = \cos(90^\circ - \theta)$.



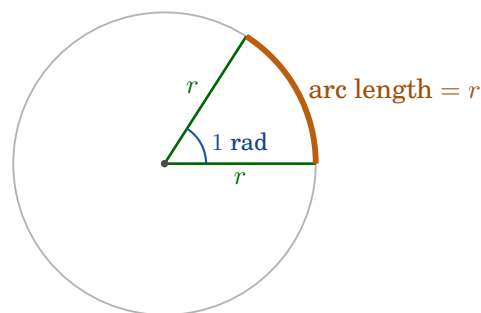
C. Radians: measuring an angle in radii

Take the radius of a circle and bend it round the edge. The angle it covers is **one radian**.

$$\theta \text{ (in radians)} = \frac{\text{arc length } s}{\text{radius } r} \quad \text{so} \quad s = r\theta$$

The whole circumference is $2\pi r$, which is $2\pi \approx 6.28$ radii laid end to end. So a full turn is 2π radians, and

$$1 \text{ rad} = \frac{360^\circ}{2\pi} \approx 57.3^\circ$$



degrees	30°	45°	60°	90°	180°	360°
radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

To convert, remember $180^\circ = \pi \text{ rad}$: multiply degrees by $\frac{\pi}{180}$, and radians by $\frac{180}{\pi}$. **Check:**

$$90^\circ \times \frac{\pi}{180} = \frac{\pi}{2} \approx 1.57.$$

Today's question: how does something going round a circle make an oscillation, and what does ω measure?

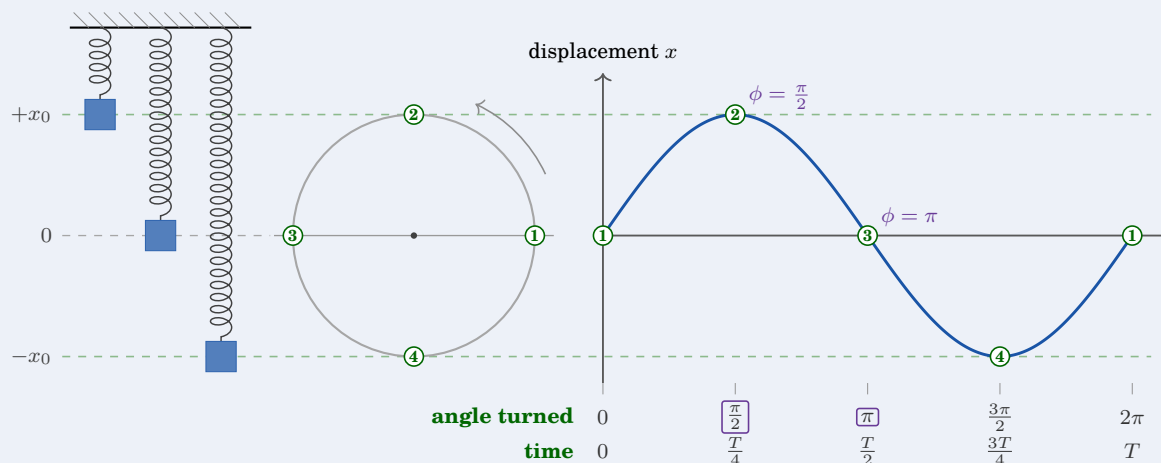
Before you start

Set your calculator to **radians** and leave it there for the whole sheet. **Check:** $\sin(\pi/2)$ gives exactly 1, and $\sin(1.57)$ gives 1.000.

Period, frequency and ω

One turn of the circle is one oscillation

Draw a circle whose radius is the amplitude x_0 , beside the spring. As the point goes round, its height follows the mass exactly: up to $+x_0$, back through 0, down to $-x_0$ and back again. The numbers show the same moment in all three pictures.



One full turn

$= 2\pi$ radians = **one complete oscillation**, from ① round to ① again

Period T , in s

the time for one full turn, so the time for one oscillation

Frequency f , in Hz

oscillations per second: $f = \frac{1}{T}$

Angular frequency ω , in rad s^{-1}

radians turned per second. The point turns 2π radians in a time T , so

$$\omega = \frac{2\pi}{T} = 2\pi f$$

Phase angle ϕ (HL)

Starting position angle at time t is $\omega t + \phi$.

Start at ①, equilibrium moving up: $\phi = 0$

Start at ②, at $+x_0$: $\phi = \frac{\pi}{2}$

Start at ③, equilibrium moving down: $\phi = \pi$

Start at ④, at $-x_0$: $\phi = \frac{3\pi}{2}$, the same as $-\frac{\pi}{2}$

Example. A mass bounces with a period of 0.50 s. Then $f = 1/0.50 = 2.0 \text{ Hz}$, and $\omega = 2\pi/0.50 = 4\pi \approx 12.6 \text{ rad/s}$: the point goes round twice a second, turning 12.6 radians every second.

Angular speed and angular frequency: two ideas, one symbol

	Angular speed ω	Angular frequency ω
Describes	something that is actually turning : a wheel, a record, a planet	something that repeats : a spring, a pendulum. Nothing has to turn
Means	the angle the object turns through per second	how many radians of the cycle pass per second, with 2π for each oscillation
Equations	$\omega = \frac{\Delta\theta}{\Delta t}$ $v = \omega r$, so $\omega = \frac{v}{r}$	$\omega = \frac{2\pi}{T} = 2\pi f$
Unit	rad/s	rad/s
Topic	A.2 Circular motion	C.1 Simple harmonic motion

Why they share a symbol. An oscillation behaves like the shadow of a point going round a circle. The oscillation's **angular frequency** is the **angular speed** of that point.

Worked example — angular speed of a record

An LP turns at $33\frac{1}{3}$ revolutions per minute.

$$f = \frac{33.33}{60} = 0.556 \text{ Hz} \quad T = \frac{1}{f} = 1.80 \text{ s} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{1.80} = 3.49 \text{ rad/s}$$

In 0.50 s it turns through $\Delta\theta = \omega \Delta t = 3.49 \times 0.50 = 1.75 \text{ rad}$.

Check: 0.50 s is a little over a quarter of the period, and 1.75 rad is a little over a quarter turn ($\pi/2 = 1.57$).

- 1) **Calculate** the angular speed of a bicycle wheel that makes one full turn every 0.40 s, and the angle it turns through in 0.10 s.

[3]

- 2) **Calculate** the angular speed of a single played at 45 revolutions per minute.

[2]

From the circle to the SHM equations

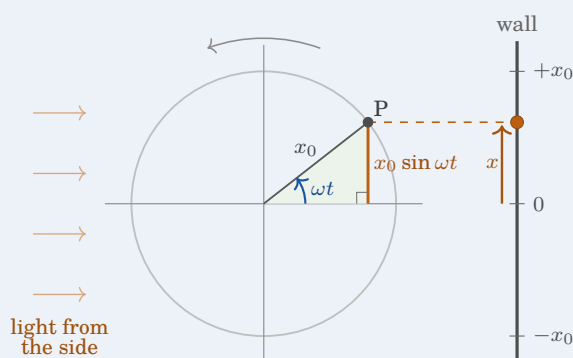
Each of the three equations below comes from the circle in a few steps. $x = x_0 \sin \omega t$ and $v = \omega x_0 \cos \omega t$ are **HL** (C1-9); $a = -\omega^2 x$ is SL, for everyone (C1-2).

1. Displacement

1. From page 2, box B: a point on a circle is at height $y = r \sin \theta$.
2. The radius is the amplitude: $r \rightarrow x_0$.
3. The angle grows steadily, ω radians every second: $\theta \rightarrow \omega t$.
4. The height of the point is the displacement: $y \rightarrow x$.

$$x = x_0 \sin \omega t$$

Check: at $t = 0$, $x = 0$; at $\omega t = \pi/2$, $x = +x_0$. These are points ① and ② on page 3.



2. Velocity

1. Once round any circle is its circumference, $2\pi r$.
2. Here the radius is the amplitude, $r = x_0$, so once round is $2\pi x_0$.
3. The point goes once round in one period T , so

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{2\pi x_0}{T}$$

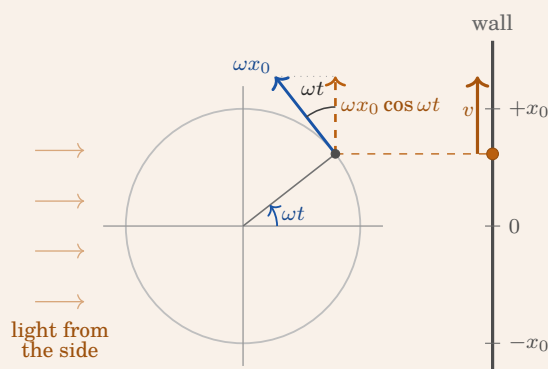
4. From page 3, $\omega = \frac{2\pi}{T}$, so

$$\text{speed} = \frac{2\pi}{T} x_0 = \omega x_0$$

5. It moves along the circle, at right angles to the radius, so its velocity makes the angle ωt with the vertical.
6. The oscillation moves only up and down, so it takes the vertical part.

$$v = \omega x_0 \cos \omega t$$

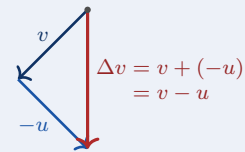
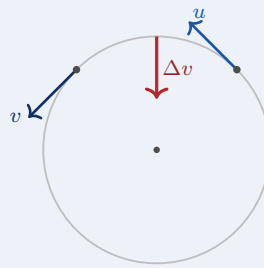
Check: at $t = 0$, $\cos 0 = 1$, so it passes equilibrium at its fastest.



Why the acceleration points at the centre

Acceleration is the change in velocity per second:

$$a = \frac{\Delta v}{\Delta t} \quad \Delta v = v - u$$



3. Acceleration

1. The velocity arrow, of length ωx_0 , also turns once every period. Its tip goes round a circle of radius ωx_0 , so, as in box 2, it changes at

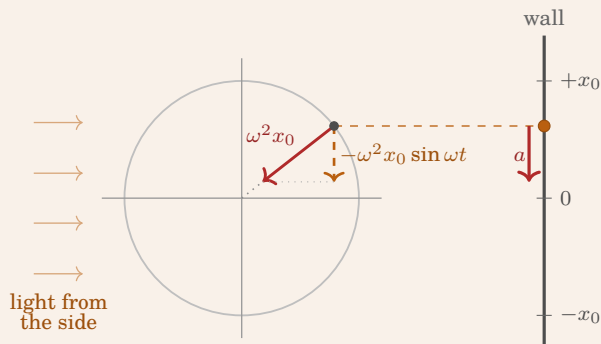
$$\frac{2\pi(\omega x_0)}{T} = \omega^2 x_0$$

pointing at the centre.

2. Take the vertical part. It points down while the point is above the centre, hence the minus sign: $a = -\omega^2 x_0 \sin \omega t$.
3. $x_0 \sin \omega t$ is the displacement x .

$$a = -\omega^2 x$$

The defining equation of SHM.



Given in your IB data booklet

You do not need to memorise these. You need to know where they come from, which is this sheet, and how to use them.

SL

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$a = -\omega^2 x$$

Basic circular motion equations

Spring-mass oscillators

Pendulum oscillators

The defining SHM relationship, $a \propto -x$

HL

$$x = x_0 \sin(\omega t + \phi)$$

$$v = \omega x_0 \cos(\omega t + \phi)$$

$$v = \pm \omega \sqrt{x_0^2 - x^2}$$

$$E_T = \frac{1}{2} m \omega^2 x_0^2$$

$$E_p = \frac{1}{2} m \omega^2 x^2$$

Displacement at any time

Velocity at any time

Velocity at any displacement

Total energy of the oscillator

Potential energy at displacement x

ω for a spring and a pendulum

Put the two expressions for the period equal to each other:

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \quad \Rightarrow \quad \boxed{\omega = \sqrt{\frac{k}{m}}} \qquad \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}} \quad \Rightarrow \quad \boxed{\omega = \sqrt{\frac{g}{l}}}$$

When a question gives the resultant force as $F = -(\text{constant}) \times x$, that constant does the job of k . **Example.** A 0.50 kg trolley has $F = -50x$: then $k = 50 \text{ N/m}$, $\omega = \sqrt{50/0.50} = 10 \text{ rad/s}$ and $T = 2\pi/10 = 0.63 \text{ s}$.